CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 7: Ratio and Proportion

Topic-wise short notes with solved examples

A ratio compares two quantities and a proportion says that two ratios are equal. This chapter covers the special kinds of ratio, continued proportion and the properties of proportion used to solve equations.

1 Ratio

- The **ratio** of a to b is $\frac{a}{b}$, written $a : b$, where $b \neq 0$. a is the **antecedent** and b the **consequent**.
- A ratio has no unit. Both quantities must be of the same kind and in the same unit.
- A ratio is unchanged when both terms are multiplied or divided by the same non-zero number.

Ratio	For $a : b$ it is
Duplicate ratio	$a^2 : b^2$
Triplicate ratio	$a^3 : b^3$
Sub-duplicate ratio	$\sqrt{a} : \sqrt{b}$
Sub-triplicate ratio	$\sqrt[3]{a} : \sqrt[3]{b}$
Reciprocal ratio	$b : a$

- The **compound ratio** of $a : b$ and $c : d$ is $ac : bd$.

Example 1.1 If $a : b = 3 : 5$, find $(5a + 2b) : (3a + 4b)$.

SOLUTION

Let $a = 3k$ and $b = 5k$.

$$\frac{5a + 2b}{3a + 4b} = \frac{15k + 10k}{9k + 20k} = \frac{25k}{29k}$$

$25 : 29$

Example 1.2 Find the duplicate ratio of $2 : 3$ and the compound ratio of $2 : 3$ and $9 : 4$.

SOLUTION

Duplicate ratio $= 2^2 : 3^2 = 4 : 9$

Compound ratio $= (2 \times 9) : (3 \times 4) = 18 : 12$

$4 : 9$ **and** $3 : 2$

2 Proportion

- Four quantities a, b, c, d are in **proportion** if $a : b = c : d$, written $a : b :: c : d$.
- a and d are the **extremes** and b and c the **means**.

$$\text{Product of extremes} = \text{product of means}: \quad ad = bc$$

- d is the **fourth proportional** to a, b and c.

Example 2.1 Find the fourth proportional to 3, 5 and 12.

SOLUTION

Let it be x. Then $3 : 5 = 12 : x$, so $3x = 60$.

$x = 20$

Example 2.2 What number must be added to each of 2, 4, 6 and 10 so that the results are in proportion?

SOLUTION

Let the number be x. Then $(2 + x)(10 + x) = (4 + x)(6 + x)$.

$20 + 12x + x^2 = 24 + 10x + x^2$, so $2x = 4$.

The number is 2. (Check: 4, 6, 8, 12 and $4 \times 12 = 6 \times 8$.)

3 Continued proportion

- a, b, c are in **continued proportion** if $a : b = b : c$.

$$b^2 = ac$$

- b is the **mean proportional** between a and c: $b = \sqrt{ac}$.
- c is the **third proportional** to a and b: $c = \dfrac{b^2}{a}$.

Example 3.1 Find (i) the mean proportional between 4 and 9 (ii) the third proportional to 4 and 6.

SOLUTION

(i) $\sqrt{4 \times 9} = \sqrt{36} = 6$

(ii) $\dfrac{6^2}{4} = 9$

(i) 6 (ii) 9

Example 3.2 If b is the mean proportional between a and c, prove that $\frac{a^2+b^2}{b^2+c^2}=\frac{a}{c}$.

SOLUTION

$b^2=ac$

$\text{LHS}=\frac{a^2+ac}{ac+c^2}=\frac{a(a+c)}{c(a+c)}$

$=\frac{a}{c}=\text{RHS}$

4 Properties of proportion

- If $\frac{a}{b}=\frac{c}{d}$, then each of the following is true.

Property	Result
Invertendo	$\frac{b}{a}=\frac{d}{c}$
Alternendo	$\frac{a}{c}=\frac{b}{d}$
Componendo	$\frac{a+b}{b}=\frac{c+d}{d}$
Dividendo	$\frac{a-b}{b}=\frac{c-d}{d}$
Componendo and dividendo	$\frac{a+b}{a-b}=\frac{c+d}{c-d}$

Example 4.1 If $\frac{3x+4y}{3x-4y}=\frac{7}{3}$, find $x:y$.

SOLUTION

Apply componendo and dividendo: $\frac{(3x+4y)+(3x-4y)}{(3x+4y)-(3x-4y)}=\frac{7+3}{7-3}$

$\frac{6x}{8y}=\frac{10}{4}$, so $\frac{x}{y}=\frac{10}{4}\times\frac{8}{6}=\frac{10}{3}$

$x:y=10:3$

Example 4.2 Solve $\frac{\sqrt{x+5}+\sqrt{x-16}}{\sqrt{x+5}-\sqrt{x-16}}=\frac{7}{3}$.

SOLUTION

Apply componendo and dividendo: $\frac{2\sqrt{x+5}}{2\sqrt{x-16}}=\frac{7+3}{7-3}=\frac{5}{2}$

Squaring: $\frac{x+5}{x-16}=\frac{25}{4}$, so $4x+20=25x-400$ and $21x=420$

$x=20$

REMEMBER

Componendo and dividendo is the quickest method whenever one side of an equation has the form $\frac{p+q}{p-q}$.

5 The k-method for proofs

- If $\frac{a}{b}=\frac{c}{d}$, put each ratio equal to k. Then $a=bk$ and $c=dk$. Substitute in both sides of the result to be proved.
- For continued proportion $\frac{a}{b}=\frac{b}{c}=k$: $b=ck$ and $a=bk=ck^2$.

Example 5.1 If $\frac{a}{b}=\frac{c}{d}$, prove that $\frac{a^2+c^2}{b^2+d^2}=\frac{ac}{bd}$.

SOLUTION

Let $\frac{a}{b}=\frac{c}{d}=k$, so $a=bk$ and $c=dk$.

$$\text{LHS}=\frac{b^2k^2+d^2k^2}{b^2+d^2}=k^2$$

$$\text{RHS}=\frac{bk\times dk}{bd}=k^2$$

LHS = RHS

6 Common mistakes

- Writing a ratio of quantities in different units without converting.
- Confusing duplicate ($a^2:b^2$) with sub-duplicate ($\sqrt{a}:\sqrt{b}$).
- Applying componendo to one side only. Every property is applied to both sides.
- Forgetting to check the answer in an equation with square roots.

MathSetu